Sinusoidal plus Residual Model

From stochastic approximation to sinusoidal-plus-residual and harmonic-plus-stochastic analysis

Digital Signal Processing for Sound and Music · Topic 7

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Roadmap

The topic extends deterministic sinusoidal/harmonic models with noise-like components.

1 · Stochastic model

Stochastic signals
Spectral envelopes
Random-phase synthesis

2 · Residual modeling

SpR / HpR
Residual subtraction
Explicit reconstruction error

3 · Stochastic residual

SpS / HpS
Residual envelope
Compact approximation

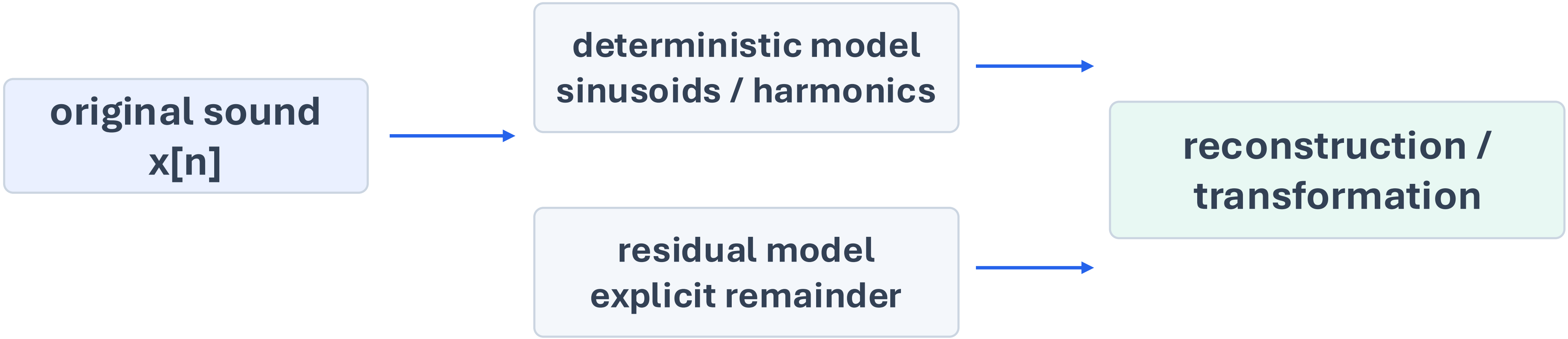
4 · Practice

Examples
Exercise 7
Inspect + listen

Why do we need stochastic or residual components?

Many sounds are not fully explained by stable sinusoids or harmonics.

The deterministic part captures stable peaks. The residual or stochastic part captures what remains: breath, attacks, turbulence, texture, and broadband noise.



Two choices for the residual: keep it explicitly (SpR / HpR), or approximate it statistically (SpS / HpS).

Stochastic signals

A stochastic signal is characterized by stable statistical descriptors, not by one exact sample sequence.

Think of it as a process, not a single curve

One observed waveform x[n] is only one realization.

Another realization may have different samples, while keeping similar energy, correlation, and spectral shape.

For this topic we model the noise-like part by matching its statistics and spectral envelope.

Two descriptors used later

Autocorrelation: similarity with delayed copies

$$R_{xx}[k] = \sum_{n=0}^{N-1} x[n] \cdot x[n+k]$$

Power spectral density: power distribution over frequency

$$P_x[k] = \lim_{N\to\infty} |X[k]|^2 / N$$

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\,2\pi kn/N}$$

In stochastic synthesis we aim to reproduce the spectral shape / envelope and perceived texture, not the exact waveform.

Stochastic model

White noise is shaped by a filter or by a short-time spectral envelope.

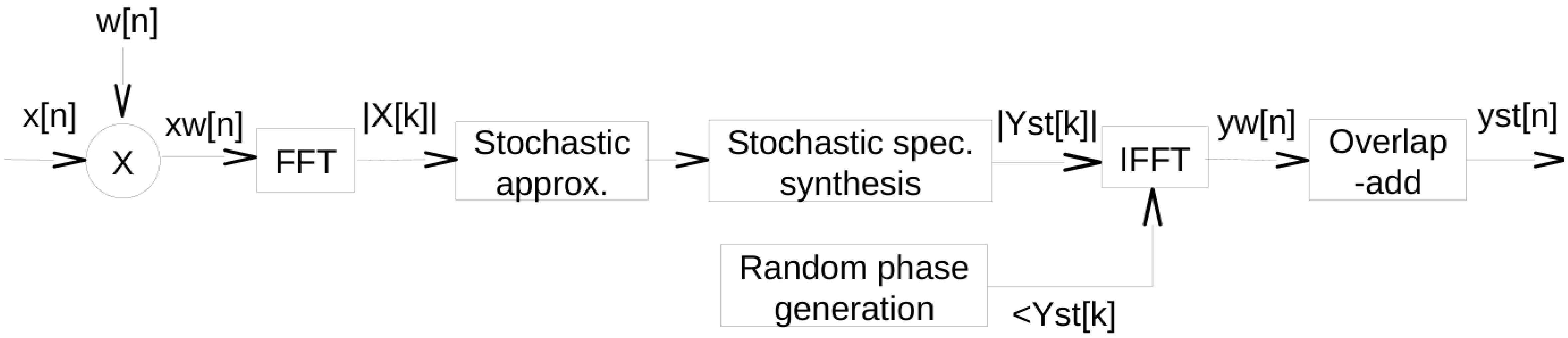
$$y_{st}[n] = \sum_{k=0}^{N-1} u[k]\, h[n-k]$$
$$Y_{st}^{(l)}[k] = |\tilde{X}^{(l)}[k]|\, e^{j\angle U[k]}$$

$u[n]$ is white noise.
$h[n]$ shapes the noise to approximate the input.
$|\tilde{X}^{l}[k]|$ is the approximated magnitude spectrum in frame l.
$\angle U[k]$ is the random phase spectrum.

The stochastic model aims to preserve the perceived noise-like texture, not a sample-perfect waveform.

Stochastic model system

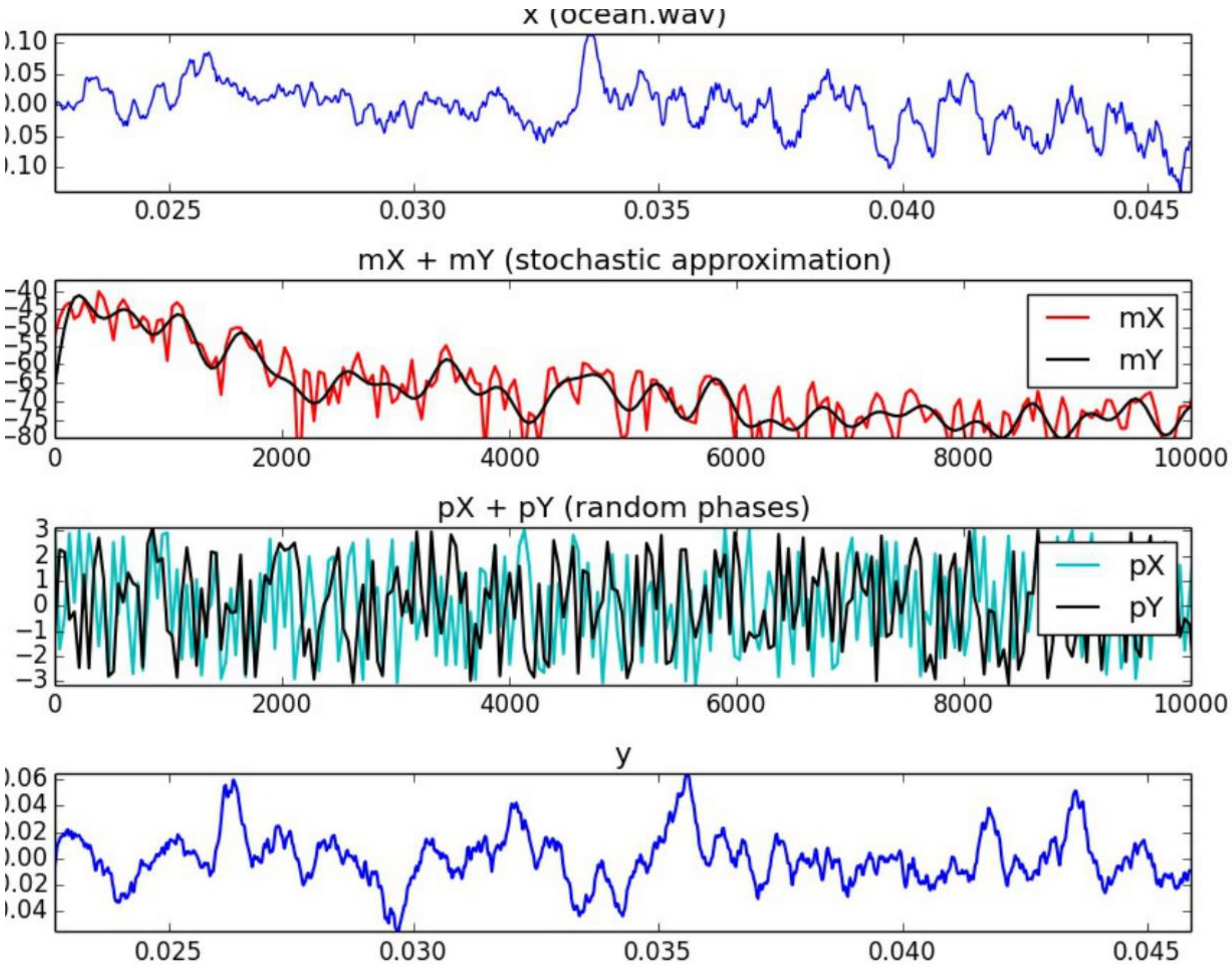
Improved block diagram of the analysis/synthesis chain.



The random-phase branch is the defining element of stochastic spectral synthesis.

Example: stochastic approximation of a noise-like sound

Large uncropped plot from the original stochastic-model material.



What to inspect

- broad magnitude envelope
- randomized phases
- similarity of texture after synthesis
- differences in exact waveform shape

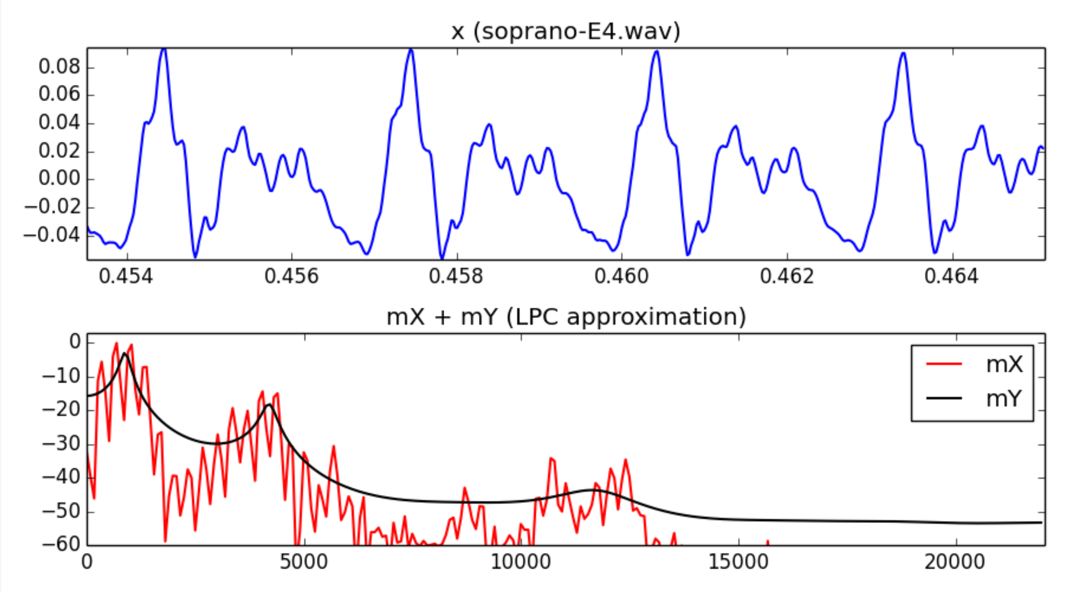
Envelope approximation

LPC and direct envelope smoothing provide compact spectral descriptions.

LPC approximation

$$\hat{x}[n] = \Sigma^{K}_{k=1}\, a_k\, x[n-k]$$

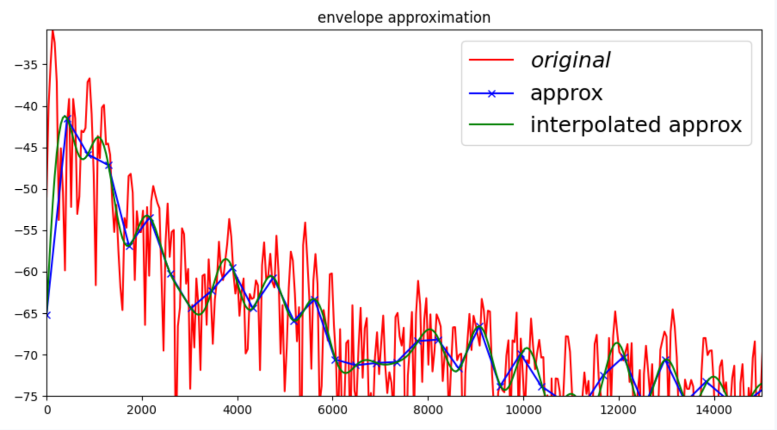
$$\text{Error} = \Sigma_n \left(x[n] - \Sigma^{K}_{k=1}\, a_k\, x[n-k]\right)^2$$

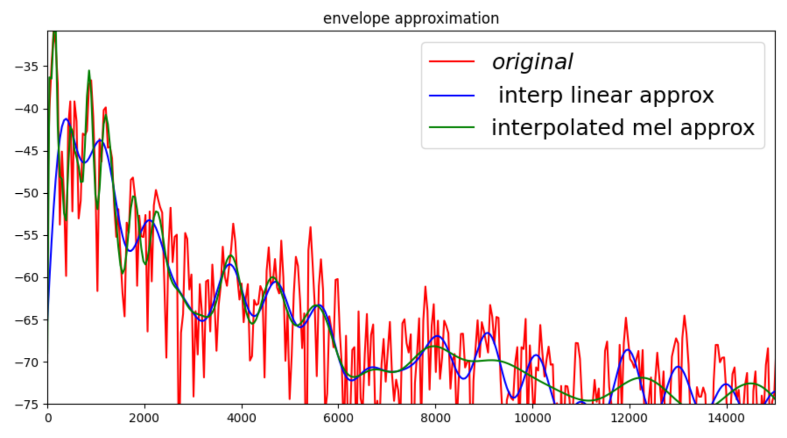


Envelope smoothing

$$\tilde{a}[k] = \text{IDFT}(\ \text{LP}(\ \text{DFT}(a[k])\ )\ )$$

$$b[k] = \text{IDFT}(\ \text{ZP}(\ \text{DFT}(\tilde{a}[k])\ )\ )$$



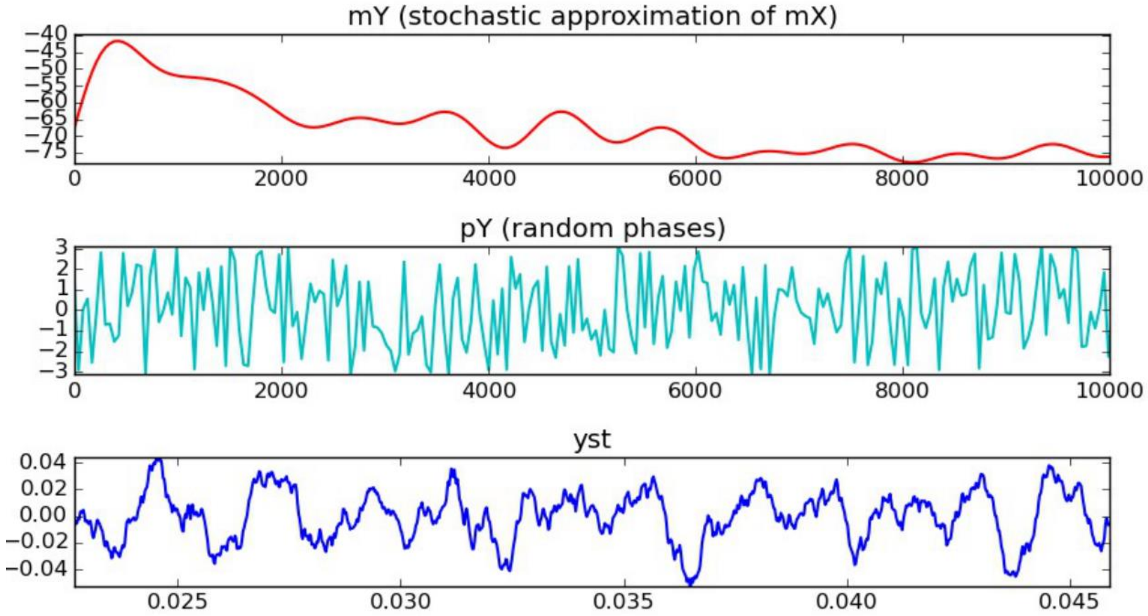


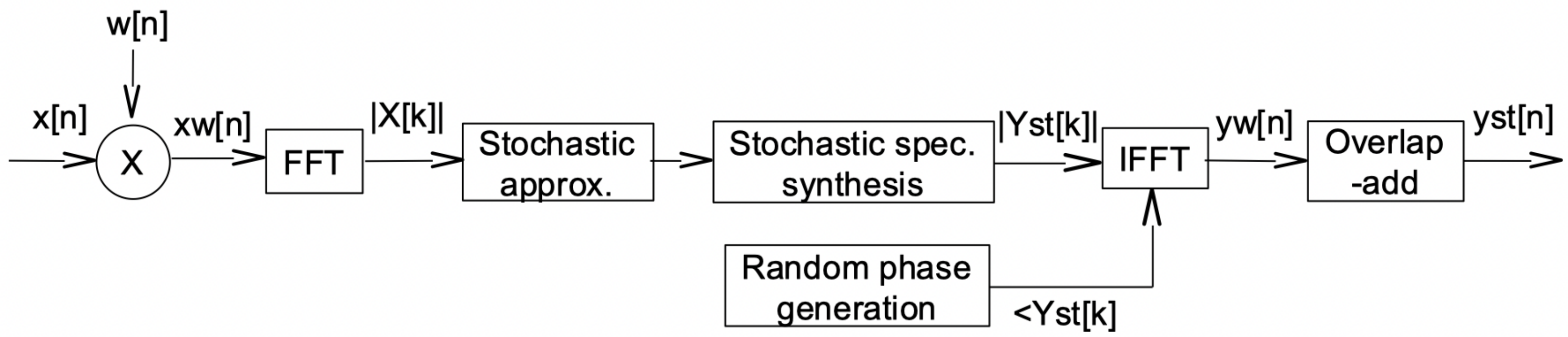
Stochastic synthesis using spectral envelopes

The synthesis keeps the magnitude envelope and generates new random phases.

$$y_{st}[n] = \mathrm{IDFT}(\ |\tilde{X}[k]|\ e^{j\angle U[k]}\ )$$

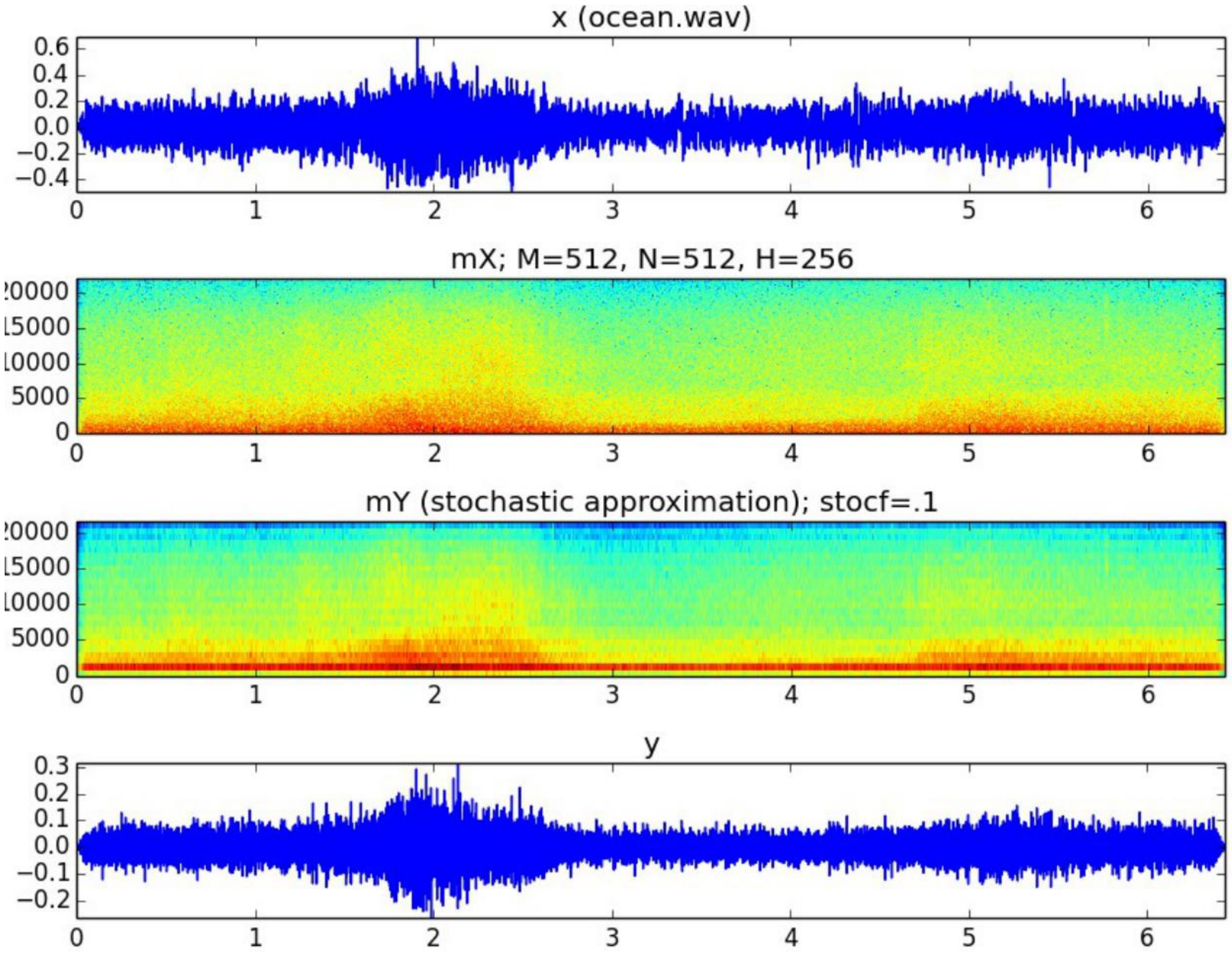
Estimate one envelope per frame.
Generate random phases.
Transform each frame back to time.
Use overlap-add to reconstruct the full signal.





Example: stochastic model over time

Compare input waveform, stochastic spectrogram, and resynthesized output.



Listening task

Decide whether the output preserves the noisy character, overall envelope, and spectral distribution of the input.

Sinusoidal plus residual model (SpR)

Keep the sinusoidal component and the residual explicitly.

$$y[n] = \Sigma^{R}_{r=1} A_r[n] \cos(2\pi f_r[n]\, n) + x_r[n]$$
$$= y_s[n] + x_r[n]$$

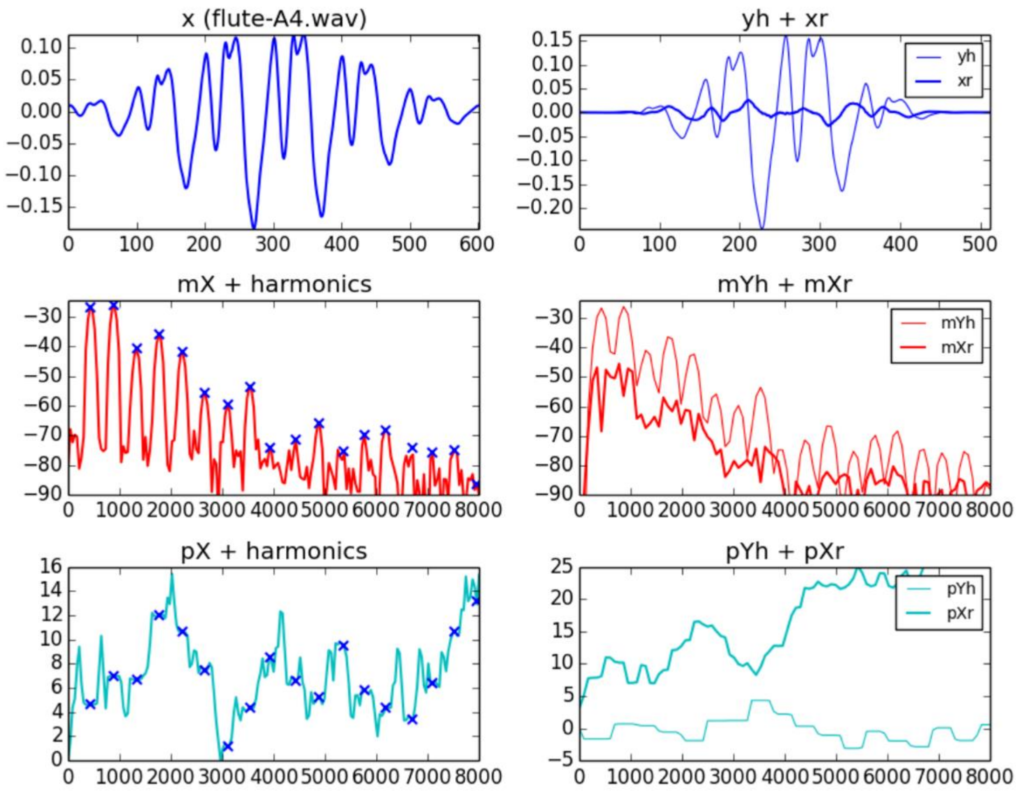
ys[n] is the sinusoidal component.
xr[n] = x[n] − ys[n] is the residual component.
The residual may carry attacks, breath, friction, or noisy components not captured by sinusoids.

SpR is appropriate when reconstruction fidelity is more important than compactness.

Spectral residual subtraction

Construct the sinusoidal spectrum and subtract it from the original spectrum.

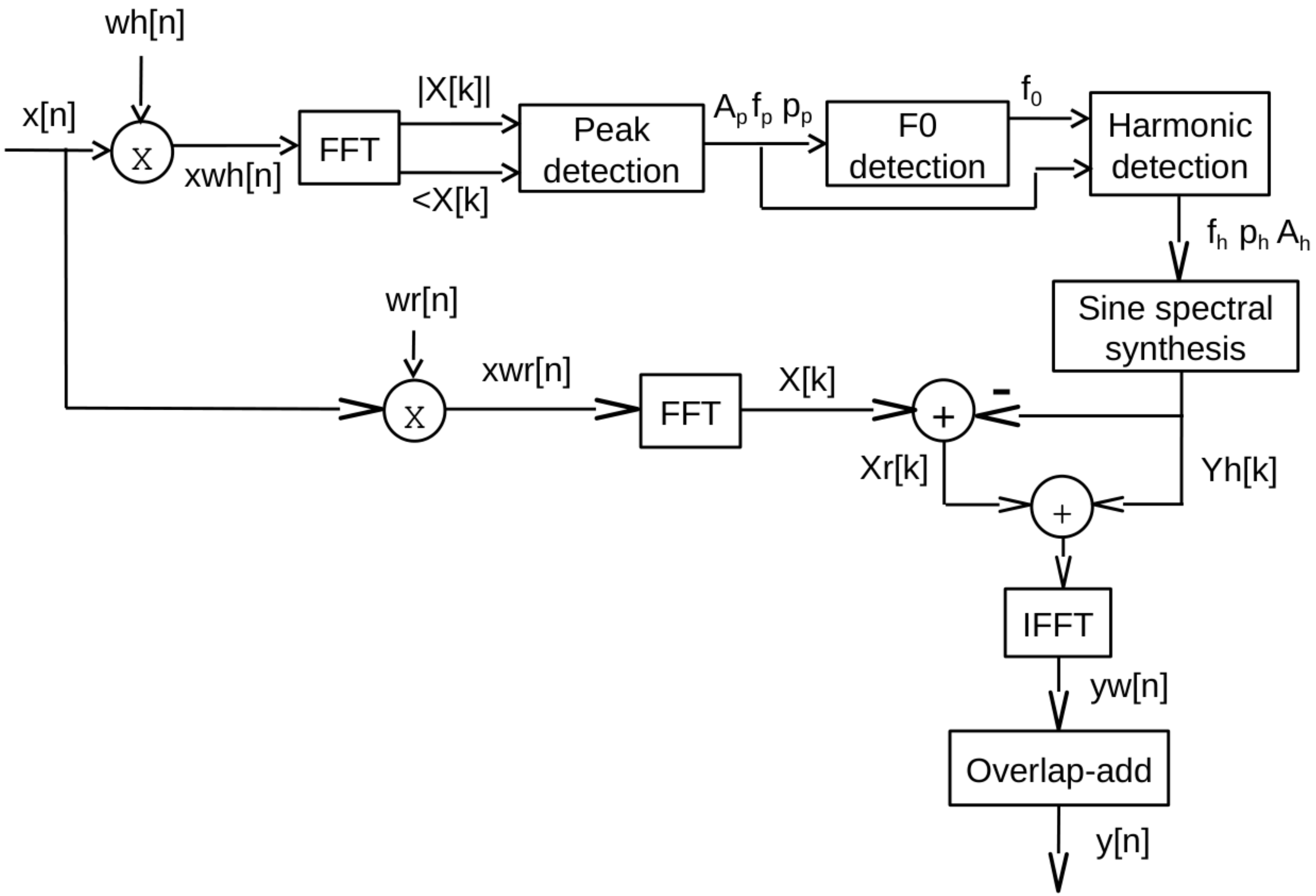
$$Y^{(l)}[k] = \sum_{r=1}^{R} A^{(r,l)}\, W[k - \hat{f}^{(r,l)}] + X_r^{(l)}[k]$$
$$X_r^{(l)}[k] = X^{(l)}[k] - Y_s^{(l)}[k]$$



- Harmonic/sinusoidal peaks are synthesized as window-shaped lobes.
- The residual is what remains after subtracting the deterministic component.
- Do not interpret the residual as “error only”: it can be musically important.

Harmonic plus residual system (HpR)

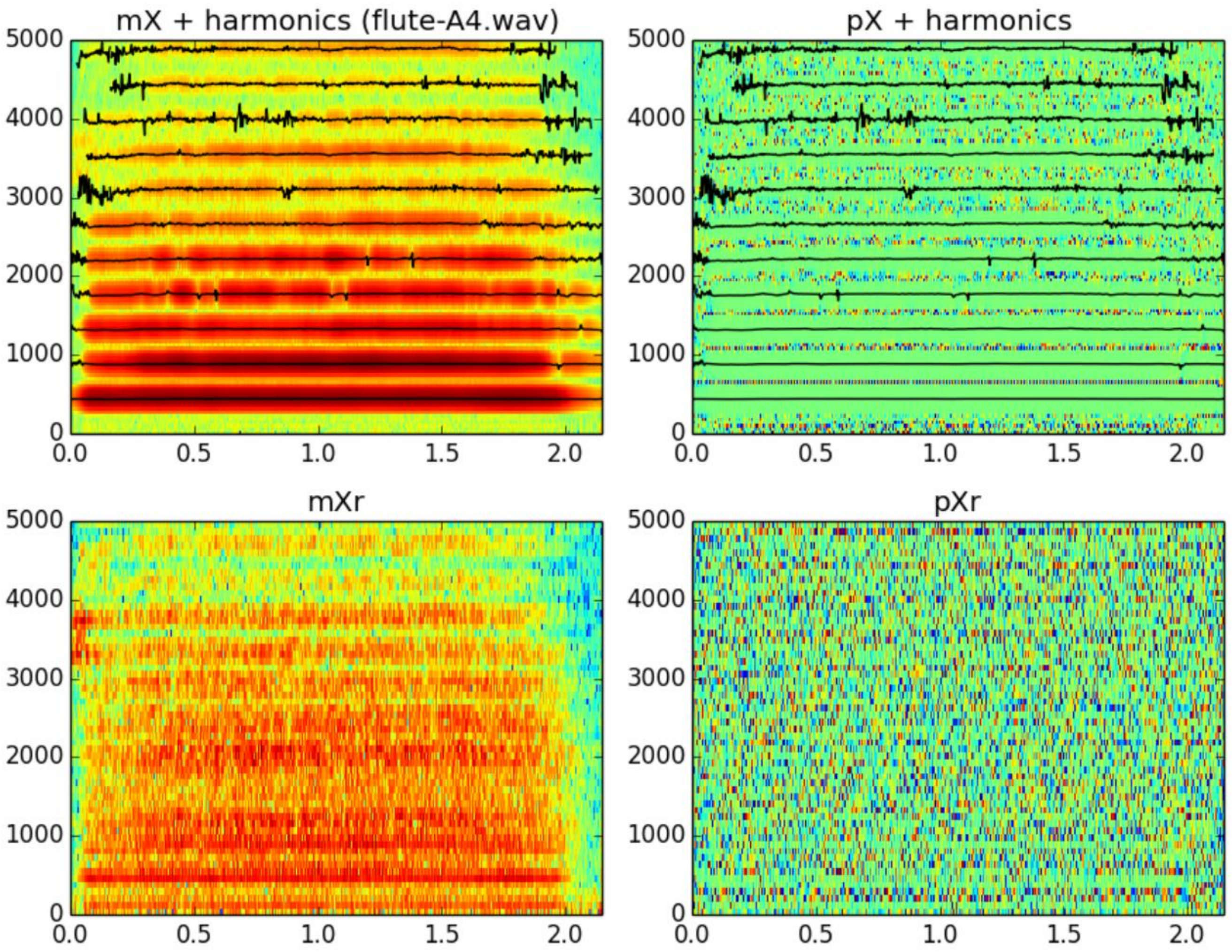
Improved diagram: harmonic analysis plus explicit residual subtraction.



HpR keeps the residual explicitly, so the lower branch carries the exact remaining spectral detail.

Example: harmonic plus residual decomposition

Large uncropped flute example: harmonics, residual magnitude, and residual phase.



How to read it

Top: original magnitude / phase plus harmonic tracks.

Bottom: residual magnitude / phase after deterministic subtraction.

Use this example to check what the harmonic model captures and what remains.

Sinusoidal / harmonic plus stochastic model

Replace the explicit residual by a stochastic approximation.

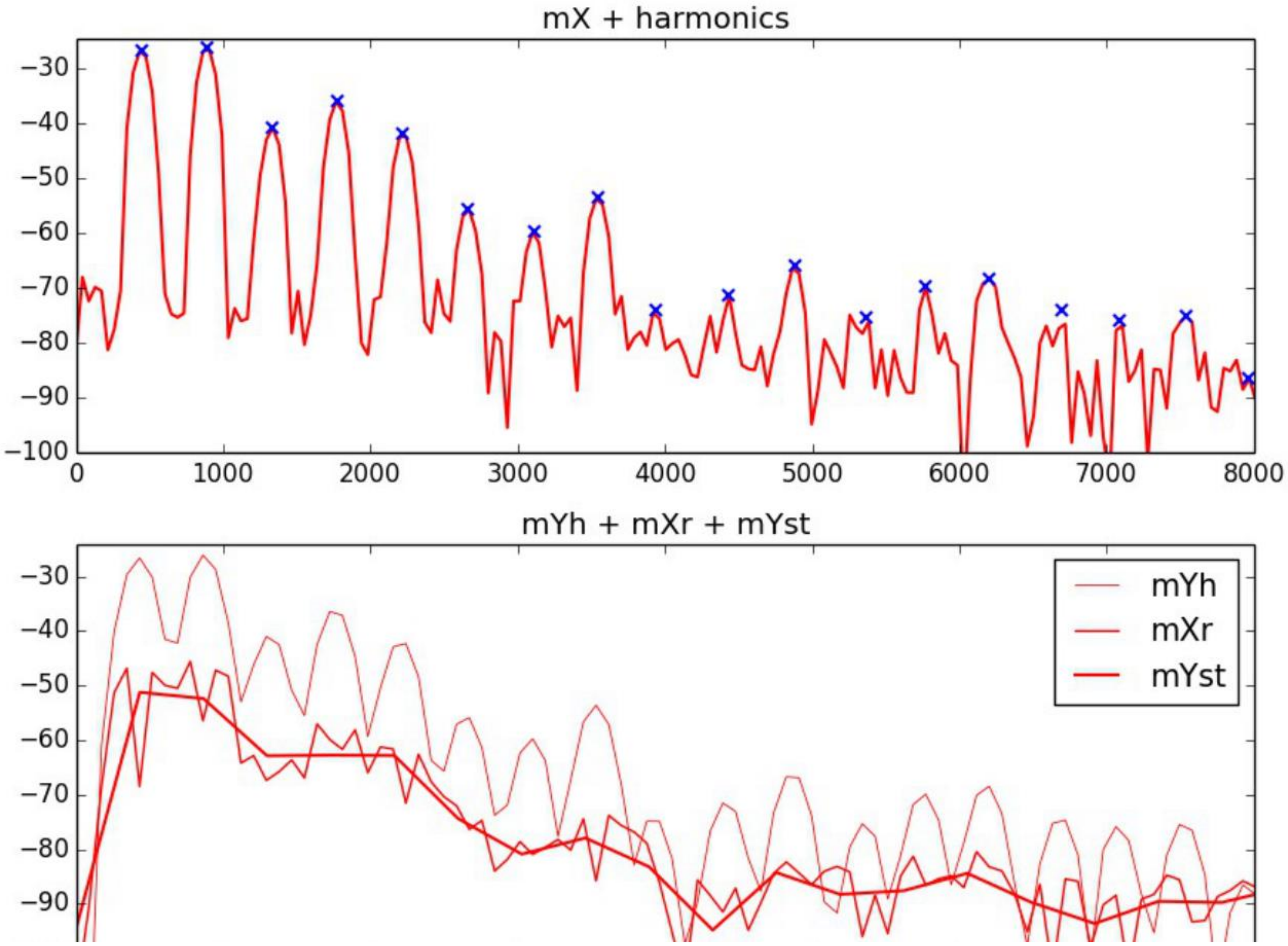
$$y[n] = \Sigma^{R}_{r=1} A_r[n] \cos(2\pi\, f_r[n]\, n) + y_{st}[n]$$
$$= y_s[n] + y_{st}[n]$$
$$Y_{st}^{(l)}[k] = |\tilde{X}_r^{(l)}[k]|\, e^{j\angle U[k]}$$

- The deterministic part explains stable peaks.
- The residual magnitude is smoothed into a stochastic envelope.
- New random phases synthesize the residual texture.

HpS / SpS trade exact residual detail for a more compact perceptual approximation.

Stochastic model of the residual

The residual spectrum is smoothed and resynthesized statistically.



What changes

• Harmonic peaks remain deterministic.

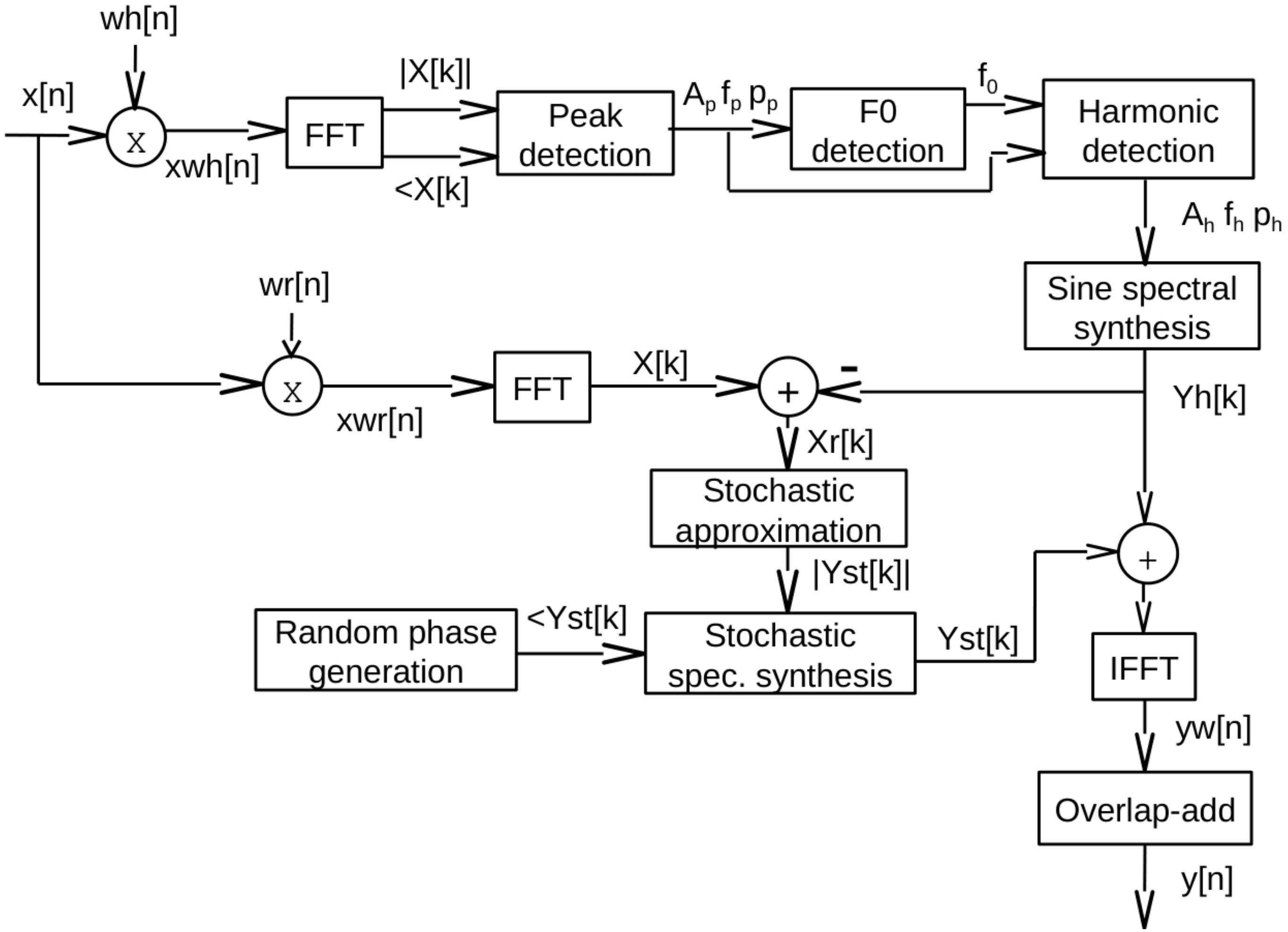
• Residual detail becomes a smooth stochastic envelope.

• Local fluctuations are not preserved exactly.

• The result is usually more compact than HpR.

Harmonic plus stochastic system (HpS)

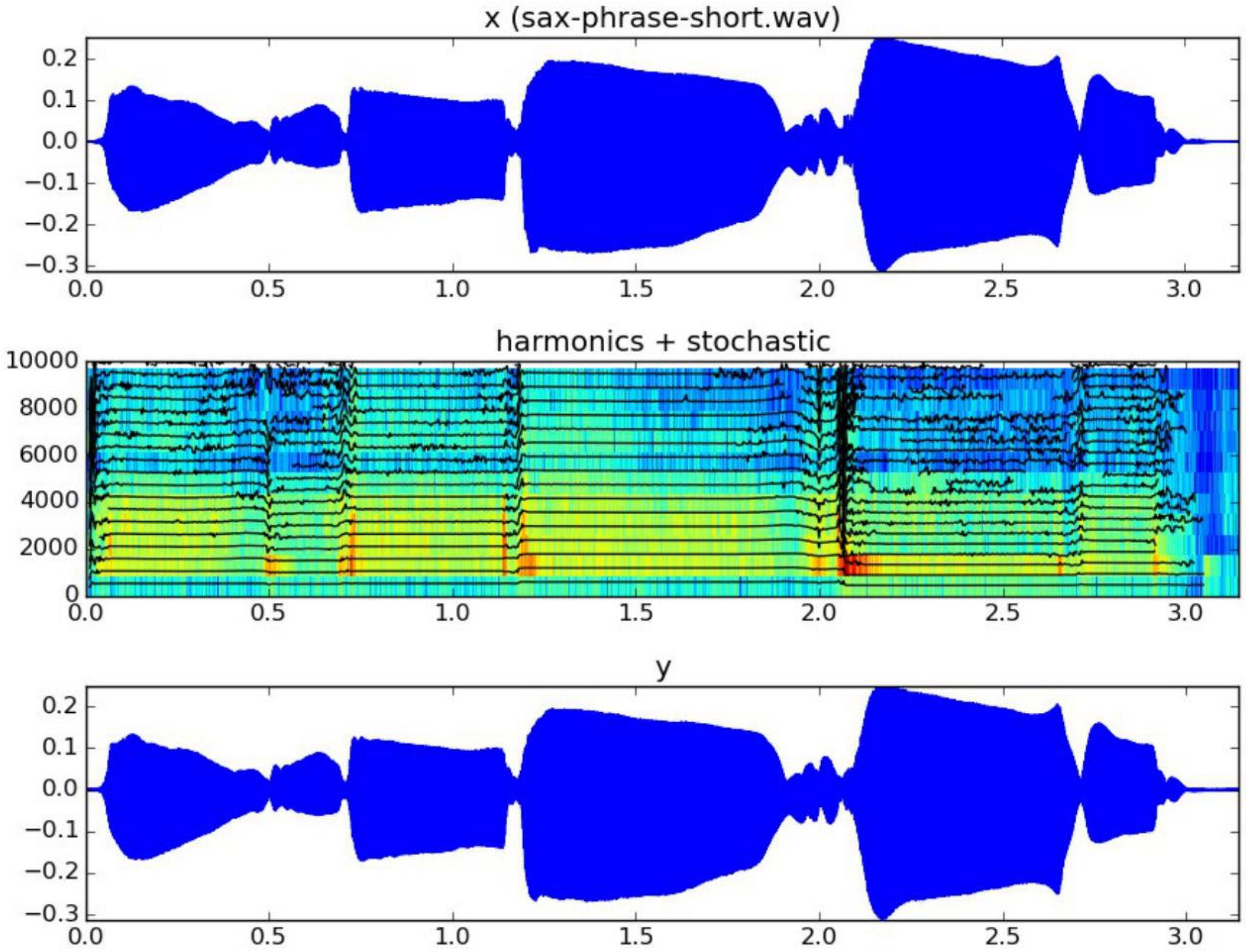
Improved diagram: harmonic model plus stochastic approximation of the residual.



HpS replaces the explicit residual by a smoothed stochastic model plus random phases.

Example: harmonics + stochastic reconstruction

A sax phrase reconstructed with harmonic tracks and stochastic residual energy.



Listen for

- pitched harmonic structure
- breath/noise component
- similarity of the output envelope
- differences caused by stochastic residual synthesis

Choosing the right model

Use the model that matches the sound and the goal of the exercise.

Model	Best for	Representation
Stochastic	Noise-like or texture-like sounds	smooth spectral envelope + random phase
SpR / HpR	Highest reconstruction fidelity	deterministic part + explicit residual
SpS / HpS	Compact perceptual approximation	deterministic part + stochastic residual envelope
Sinusoidal / Harmonic only	Clean tonal or pitched material	deterministic peaks and tracks

Examples to use with this topic

Connect each plot with a notebook, a parameter choice, and a listening outcome.

1 **stochasticModel_function.py / stochastic model examples**

inspect spectral-envelope approximation and random-phase synthesis

2 **sprModel / hprModel examples**

inspect deterministic subtraction and explicit residual reconstruction

3 **spsModel / hpsModel examples**

inspect residual smoothing and stochastic residual synthesis

Exercise 7: workflow

Use the same lab structure as the previous topics: change one parameter, inspect, listen.

- Choose one mostly noisy sound and one pitched sound with clear harmonic structure.
- Run a stochastic-model analysis/synthesis on the noisy sound and compare spectral envelope and output audio.
- Run a harmonic-plus-residual model on the pitched sound and inspect harmonic tracks, residual magnitude, and reconstruction.
- Run a harmonic-plus-stochastic model and compare it with the residual version.
- Write short comments that connect parameters, plots, and listening results.

Exercise 7: parameter experiments

Change one parameter at a time and explain the visible and audible consequence.

Parameter	What to observe
analysis window	resolution, smoothing, transient detail
threshold	which deterministic peaks are kept before residual subtraction
envelope smoothing	how detailed or smooth the stochastic residual becomes
stochastic scaling	how much noisy component is present in the output
number of harmonics	which deterministic components remain outside the residual

What to remember from Topic 7

Stochastic

Match a smooth magnitude envelope and use random phases.

Residual

Keep what the deterministic model does not explain.

SpR / HpR

Use explicit residuals when fidelity matters.

SpS / HpS

Use stochastic residuals when compact perceptual approximation is enough.

Validation

Always compare plots and audio together.

References and credits

- Original slides: “7T1: Stochastic Model” and “7T2: Sinusoidal plus Residual Modeling”.
- sms-tools models: stochasticModel.py, sprModel.py, spsModel.py, hprModel.py, and hpsModel.py.
- sms-tools-materials: examples, exercises, lecture plots, and sound files for classroom use.
- Use the examples and Exercise 7 to connect the equations, plots, parameters, and listening results.